

Multiscale Modeling of Biological Systems: Neuroscience

Assignment 4

Group 12

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1 Introduction

Deep neural networks trained on sound recognition are a popular class of models of the auditory cortex, but whether architectural choices make a network act more like a brain or just act more accurate is a deeper question. We tried testing this with representational similarity analysis between network layers and fMRI responses from the superior temporal gyrus, a region of the brain thought to hold immediate acoustic to semantic representative sounds.

1.1 Data, purpose and models

The superior temporal gyrus (STG) is the region of the auditory cortex that represents sound somewhere in between their acoustic features and their meaning. The goal of this study was to find out whether models that are designed to be more like the auditory system also represent sounds more similar to how the STG does and not just classify them better.

The brain data provided for this assignment comes from the fMRI dataset of Santoro. It has the STG responses (the beta values) of one participant to 288 natural sounds from about 73 categories. We compared this data to three small models that were trained on ESC-50, which is a dataset of 50 environmental sound classes with close to 1600 training clips. Each model adds more auditory structure than the previous / The waveform model is a simple MLP on the raw audio, the uninspired model is a CNN on mel spectrograms and the inspired model adds a recurrent GRU layer and uses ReLU6 activations. We ran five trained runs of each model and also included YAMNet which is a much bigger model trained on AudioSet for reference as it is seen to be aligned well with the STG previously.

Table 1: Overview of the compared models.

Model	Input	Conv.	Recurrent	Nonlinearity	Parameters
Waveform	raw waveform	×	×	Tanh	8,893,554
Uninspired	mel spectrogram	✓	×	Tanh	43,122
Inspired	mel spectrogram	✓	GRU	ReLU6	203,314
YAMNet	log-mel spectrogram	✓	×	ReLU	~3.7M

1.2 Study type

This is primarily a contrastive study, in this type of study, we don't necessarily ask what the optimal solution to a task is, or fit a model to the brain, what we try instead, is to compare several model classes that differ in hypothesis about auditory processing. The architectures also are constructive in a way, since each adds one biologically motivated component.

2 Methods

For every sound we obtained a feature vector, and for every pair of sounds the correlation distance $1 - r$ (Pearson) between vectors, giving a 288×288 RDM with sounds in identical order for all RDMs. The brain RDM was computed on the STG voxel patterns (`SantoroDataset().brain_responses`). For the networks we used `extract_activations`, which returns the output of every trainable layer (after the layer’s nonlinearity when one follows, before pooling; the classifier returns logits) and flattens it per sound. This gives six layers for the waveform and uninspired models and five for the inspired model, where the two stacked GRU layers count as one `rnn` layer (output of the last GRU layer). RDMs were built with `rsatoolbox`.

2.1 RSA analysis

Before getting the activations, all 288 sounds were resampled to 8 kHz and cut to their first second, so the input matches what the model was trained on. For each architecture we made RDMs for the untrained model with random weights and we kept the best checkpoint of each of the five trained runs so we could compare them later. For YAMNet we used the embedding layer.

To measure how well a model aligns with the STG, we compared its RSM to the STG RDM using only the upper triangle. This is because the matrix is symmetric and the diagonal is always 0. The main metric we used was the Spearman rank correlation, as it only assumes that the two RDMs have the same ordering of distances and no linear relationship between them.

2.2 Statistical tests across multiple models per type

As we have five runs of each model, we show every alignment score as the mean over the runs with a 95% confidence interval based on the t-distribution with four degrees of freedom. This shows us how much the result depends on the random initialisation but not how it changes between people as the brain data comes from one person.

For the main comparisons we used the final layer of each model, we compared trained and untrained versions of each architecture and each pair of trained architecture which gives us six planned tests. Because the untrained models are not the starting points of the trained ones, we treated all groups as though they were independent and used Welch’s t-test. We then corrected the six p values with Holm–Bonferroni method to limit the number of false positives. We however did not test the differences between layers.

2.3 t-SNE visualisation

To see how layers organise sounds, we embedded the layer activations of one trained run per architecture (and the brain patterns and YAMNet) in 2D with t-SNE (perplexity 30, fixed seed) and coloured points by sound category.

3 Results

3.1 Effect of training

Training did not clearly change how well the final layer agreed with the STG. For waveform the mean score increased slightly from -0.048 to -0.04, and it fell for uninspired (from 0.052 to 0.029) and for inspired (from 0.074 to 0.055). None of these changes were significant after Holm correction (all $p > 0.05$), however uninspired got very close with $p = 0.060$. The trained runs of uninspired and inspired varied less than the untrained ones as the smaller confidence intervals show.

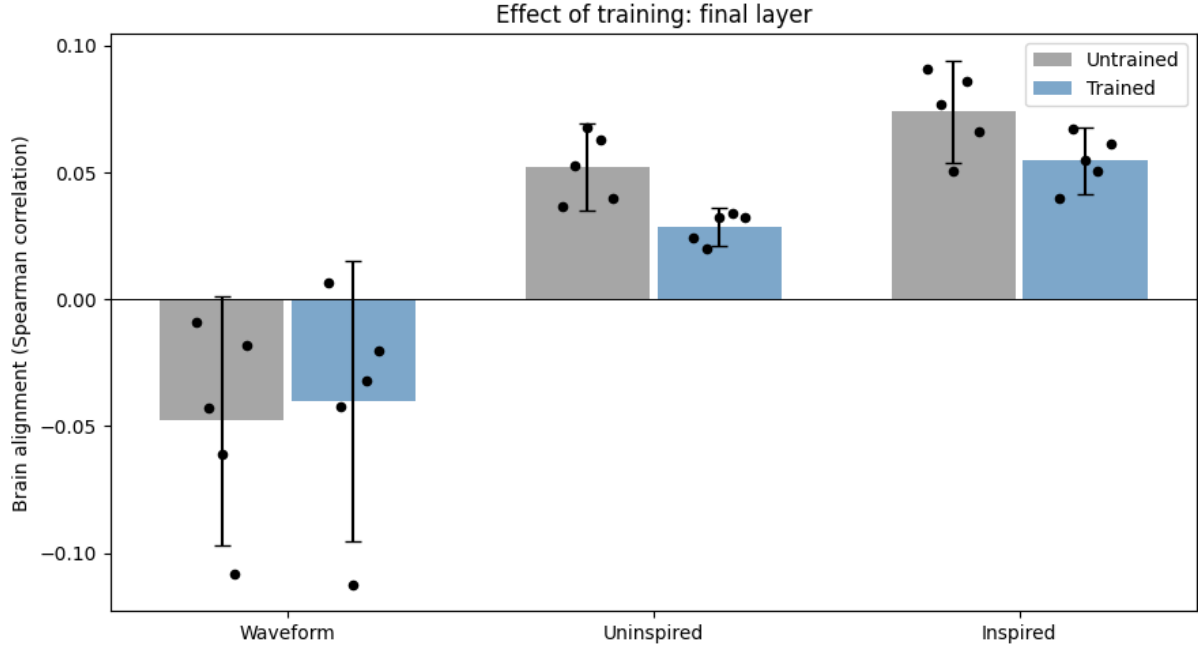


Figure 1: Final layer alignment with the STG for untrained and trained models. Bars show the mean over five runs.

3.2 Effect of architecture

Within the trained models, Inspired had the highest final-layer alignment (0.055), then uninspired (0.029) then waveform (-0.0040). The corrected tests showed that inspired aligned better than both uninspired and waveform, but there was no clear difference between uninspired and waveform. YAMNet’s embedding scored 0.028, roughly the same as uninspired and below inspired. All scores were not large, so even the best model only weakly agrees with the brain data.

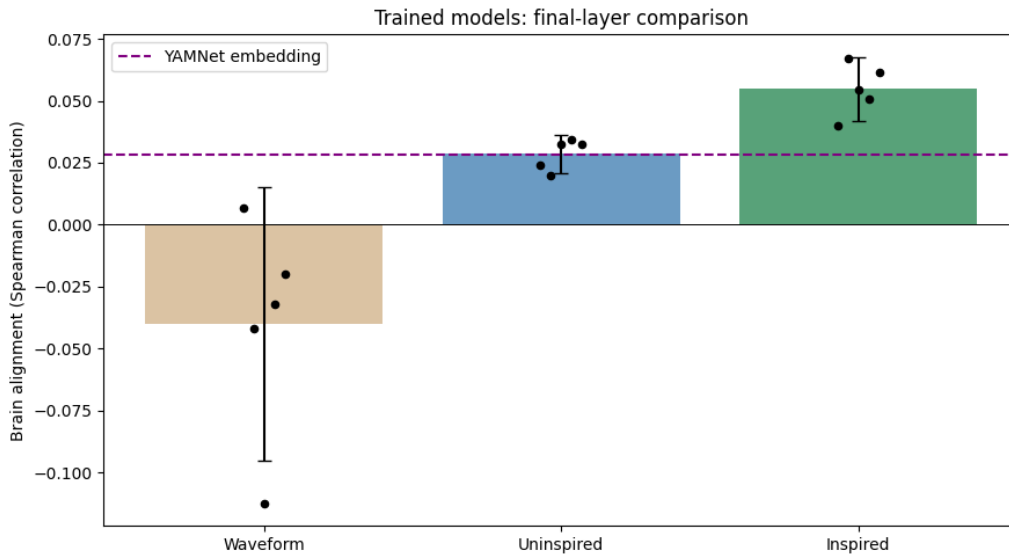


Figure 2: Final-layer alignment of the trained models. The dashed line shows the YAMNet embedding as a reference (single model, no interval).

Table 2: Final-layer alignment with the STG (Spearman ρ , mean over five runs with 95% CI) and Holm-corrected p -values of Welch’s t -tests.

Model	Untrained ρ	Trained ρ	Change	p (training)
Waveform	-0.048 ± 0.05	-0.040 ± 0.055	+0.008	0.778
Uninspired	0.052 ± 0.017	0.029 ± 0.008	-0.023	0.060
Inspired	0.074 ± 0.020	0.055 ± 0.013	-0.019	0.121
YAMNet	—	0.028	—	—
Trained Waveform vs. Uninspired				0.076
Trained Waveform vs. Inspired				0.038
Trained Uninspired vs. Inspired				0.015

3.3 Effect of layer depth

We compared every extracted layer of each trained architecture with the STG RDM using Spearman correlation, averaged over runs with 95% confidence intervals. In all three trained models, the final layer aligned better with STG than the first layer. However, the relationship was not monotonic: for waveform and uninspired, the highest average score happened to be at an intermediate layer rather than at the output. Inspired’s final average (0.055) was the highest among trained models. Because all scores are small and come from a single participant, we didn’t test layer differences statistically and interpret these trends in too much depth.

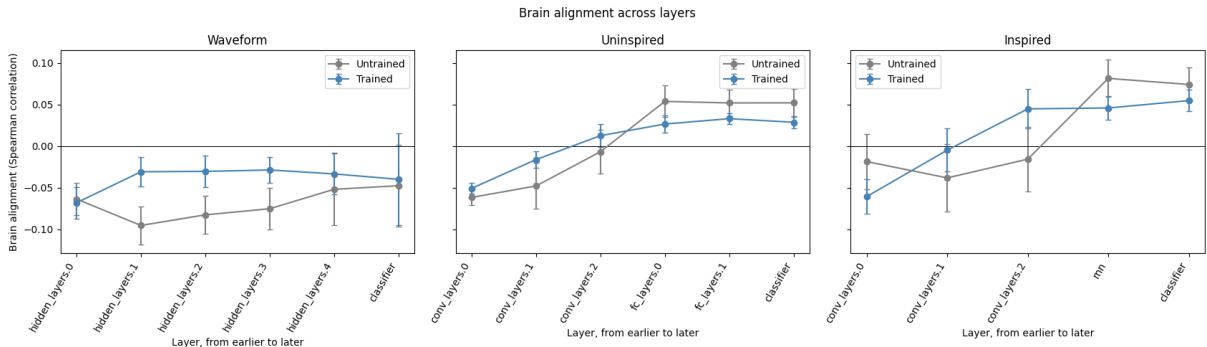


Figure 3: Layer-wise alignment with the STG for untrained (grey) and trained (blue) models. Mean over five runs with 95% CI.

3.4 t-SNE visualisation

The t-SNE embeddings of the STG brain data were compared with YAMNet embeddings and model layers. None of the methods were able to fully separate the categories of each sound into local neighbourhoods. Uninspired and inspired models were able to gather same-category points more closely compared to waveform model. The YAMNet embeddings were the most compact and accurate in terms of separation. In general, the model and YAMNet representation t-SNE embeddings were more concentrated compared to the STG brain responses (see Fig. 4).

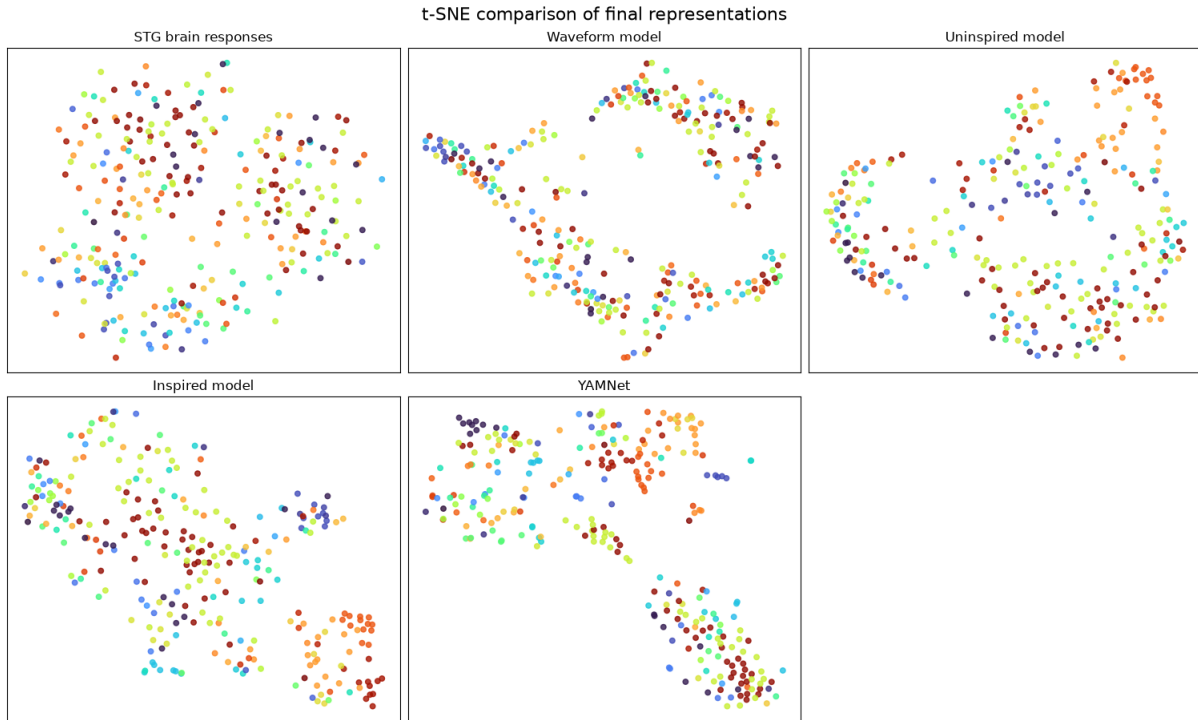


Figure 4: t-SNE projection of the STG brain data, last layer activations in waveform, inspired, uninspired models and YAMNet embeddings.

4 Discussion

4.1 Untrained versus trained models

The untrained uninspired and inspired models already aligned with the STG above zero and the training didn't improve this. This could be because most of their alignment comes from architecture and the spectrogram input, not from what they learned. A mel spectrogram already splits sound into frequency bands, so even the random filters on top of it keep some brain-like structure. Training may have even lowered alignment slightly because the final layer gives the class scores for the 50 ESC-50 classes. These classes don't match the categories of the Santoro sounds, and the STG could hold intermediate rather than purely category-level information. With only 1600 training clips the models also could have overfitted to ESC-50. Training did make the runs more similar to each other, which could mean that trained models end up with similar solutions.

4.2 Comparison between models

Inspired aligned the best because it differs from uninspired in multiple ways (the GRU, ReLU6 and more params), however we can't tell which one of these caused the improvement. One possible explanation is that GRU integrates the information over time like later auditory regions similar to the STGs do. The waveform model was the worst even though it by far has the most parameters so model size alone doesn't explain the alignment. Working When we work directly on the raw waveform it makes its activations depend on the exact timing and phase of the signal, which the brain mostly ignores at this level. YAMNet only cared about the same as uninspired, even though it is much larger and trained on far more data. This could be as we only used its embedding layer which is close to its output and more task-specific compared to intermediate layers.

4.3 Role of model depth

Deeper layers did not align better with STG uniformly. The final layers did better than the first layers, consistent with later stages' encoding being more abstract, and category related structure, as expected for an intermediate to semantic region. Yet Waveform and Uninspired peaked before the output, possibly because their last layers become specialised for the 50 ESC-50 classes and discard acoustic information that STG retains. Across architectures, Inspired, the most structured model, showed the best final layer alignment, while Waveform, with almost no classification skill (about 5% validation accuracy), was negative. Depth therefore helps only when the architecture supports meaningful hierarchical transformations, and gains remain small.

4.4 Further observations

Several further patterns emerged. Task performance and brain alignment were only loosely related: Uninspired reached roughly 29% validation accuracy and Inspired about 40%, yet the alignment averages of both fell slightly after training, and corrected tests showed no clear training effect. YAMNet, a much larger network trained on far more data, scored 0.028, similar to Uninspired and below Inspired, so our small models are not obviously worse than a reference. Better classification therefore does not guarantee better alignment. Finally, all effects are small, the brain data come from one participant, and five runs per model limit statistical power, so absence of significance should not be read as absence of an effect.

4.5 Insights from t-SNE

The t-SNE maps help explain alignment scores by showing whether a representation groups sounds by semantic category, as STG is thought to do. A model whose final layer forms compact category clusters should have an RDM with low within each category and high between each category dissimilarity, which tends to raise Spearman correlation with STG. Conversely, diffuse maps, as expected for the barely trained Waveform model, should coincide with weaker alignment. However, t-SNE maps are trained independently and try to preserve local neighbourhoods, which may distort the distances between clusters. Thus, t-SNE maps do not provide a reliable indication of alignment and cannot be used alone to compare different models, as they should only support the RDM comparison.

5 Future Work

Our models are small, trained on 1600 videos, and compared against data from a single participant, so the largest gains will probably come from making architectures and training regimes more brain like. Two directions follow from the lecture material: constraining the architecture with known anatomy of the auditory pathway, and changing how the network is trained and evaluated. Both can be tested with the same RSA pipeline, including untrained baselines and multiple runs per model, which keeps comparisons statistically controlled. Ideally, future evaluations would also use brain data from several participants and additional auditory regions to test whether conclusions generalise beyond STG.

5.1 Injecting neuroanatomical detail

The auditory pathway could be mapped onto network stages more literally: cochlea like filter-bank front end, brainstem and inferior colliculus, medial geniculate thalamus, primary auditory cortex, and then belt and parabelt regions leading to STG. Each stage could be a module with realistic size, receptive field structure and tonotopic organisation. Biological connectivity would add skip connections, such as thalamic input reaching nonprimary areas, and feedback connections from higher to lower stages, implemented with recurrence. Adding parallel pathways for

spectral and temporal modulation, plus lateral inhibition, could give layers distinct roles that can be matched to specific brain regions instead of only the final layer.

5.2 Other interventions

Beyond architecture, training could be changed. Self supervised or contrastive pretraining on large audio corpora may give richer features than supervised learning on 1600 ESC-50 clips, and fine tuning could follow. Data augmentation and multitasking objectives, for example combining classification with prediction of upcoming sound, may encourage more general representations. Biologically motivated constraints such as noise, sparsity, or energy penalties could further regularise activations. Finally, models could be fitted directly to brain data, for example by adding an RDM-matching loss or fitting voxelwise encoding models, though this risks overfitting one participant and should be validated on held out sounds and participants.